

Human Natural Structure

Standards & Conformance Specification

A Technical Mapping of HNS to International AI Standards and Regulatory Obligations

Volume 6 · Human Natural Structure Series

Satoru Hara

Natural Structure Works (NSW)

A Note on Method and Status

This volume is written in the form of a technical standard — with clauses, normative references, conformance requirements, and the auxiliary verbs (SHALL, SHOULD, MAY) that standards use. The form is deliberate: it states precisely what an implementation would have to do. The form must not, however, be mistaken for status, and the reader is owed an unusually direct statement of that distinction.

This is a proposed specification, not an adopted standard. Human Natural Structure has not been reviewed or adopted by ISO, IEC, JTC 1/SC 42, CEN-CENELEC, NIST, IEEE, ITU, or any other standards body, and nothing here carries regulatory force. The normative verbs define conformance to this document; they do not impose any legal obligation. Where the text maps HNS to an article of law or a clause of a standard, it states how HNS would support that provision if adopted — a proposal placed before the standards community, not a description of an existing requirement.

Three registers. Established: the regulations and standards referenced — the EU AI Act, the ISO/IEC SC 42 family, the NIST framework, the W3C provenance stack, the functional-safety and security standards — are real, current artifacts the reader can verify. HNS design: the coordinate, the verification architecture, the control system, and their mappings to those instruments are the author’s engineering proposals. Conjecture: that HNS will be adopted, and that adoption would deliver the enforcement the framework argues for, is a wager, marked as such.

Currency. The regulatory landscape described here is current to mid-2026, including the provisional Digital Omnibus agreement of 7 May 2026 deferring the AI Act’s high-risk deadlines. That landscape moves; before any reliance, the reader should consult the Official Journal of the EU and the standards bodies directly. As elsewhere in the series, named instruments indicate the ground this volume builds on, not endorsement by their authors.

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0. Foreword

0.1 Purpose

This specification defines how Human Natural Structure (HNS) and its verification and control architectures are expressed as a conformance-checkable technical layer beneath the existing process standards and laws that govern artificial intelligence. Its purpose is to supply what those instruments require but do not themselves provide: a technical means by which conformance can be verified, recorded, and enforced at the level of the individual inference.

0.2 Background — the enforcement gap

The world's AI regulations and management standards converge on a demand — that AI be safe, logged, transparent, and subject to human oversight — and stop short of specifying the technical mechanism by which any of these is to be achieved. The EU AI Act mandates record-keeping and human oversight for high-risk systems without fixing what is recorded or how oversight is exercised; ISO/IEC 42001 certifies management processes, not inference-level behaviour; the NIST framework names a MEASURE function whose tooling is the least mature of the five. The result is a gap between requirement and implementation: the obligations exist, the technical layer that would discharge them does not. This gap is this volume's subject, and the claim — advanced as a proposal — is that the HNS stack is a candidate for that missing layer.

NOTE *The gap is now visible in the law's own timetable. Under the Digital Omnibus agreement of 7 May 2026, the application of the AI Act's high-risk obligations is linked to the availability of harmonised standards and support tools, and the Annex III deadline has been deferred to 2 December 2027. A regulation is, in effect, waiting on a technical layer that does not yet exist — the situation this specification addresses.*

0.2.1 The gap, in numbers

The gap is not only conceptual. The function most able to close it is the least developed: in the NIST framework the MEASURE function — the one that would quantify and verify behaviour — has the least mature tooling of the five. The infrastructure is correspondingly thin: industry surveys in 2025 found that fewer than one in five

organizations, even in regulated sectors such as healthcare and finance, had dedicated AI incident-reporting tooling in place. Most organizations have identified their AI risks yet lack the means to record and respond when those risks materialize. That deficit — measurement and recording at the level of the individual inference — is what this specification proposes to fill.

0.2.2 The missing-link claim, stated and bounded

The claim of this volume can now be stated exactly. Existing instruments supply the obligations; HNS proposes the inference-level technical layer — coordinate, verification, control — by which those obligations could be made checkable and enforceable. That layer is the “missing link.” The claim is bounded in two ways: it is a proposal, not an adopted requirement; and it is an enabling layer, not a guarantee — it makes conformance checkable, it does not make a system correct. With those bounds, the missing-link claim is a contribution to the standards conversation, not a substitute for it.

0.3 Relationship to international standards

This document does not replace any standard. It is designed to sit beneath ISO/IEC 42001, the EU AI Act, the NIST framework, and their peers, supplying the inference-level verification and control they assume. Clause 2 lists the instruments treated as normative references; Annex G maps the HNS components to them in detail.

0.4 Position in the HNS series

This volume connects the entire series as the final conformance layer.. It depends on, and does not restate, the core theoretical foundation of Volume 1, the coordinate system of Volume 2 and 3 (HNS-36/144/864), the grounding stack of Volume 4 (SMS-6), and the verification architecture of Volume 5 (EVA).. Those volumes are the normative source for their respective components; this volume references them and adds the standards, legal-basis, conformance, and control layer.

1. Scope

1.1 Subject matter

This document specifies: the expression of the HNS coordinate and its verification and control architectures as a conformance-checkable layer; the machine-readable record format by which conformance is evidenced; the mapping of that layer to international AI standards and to the EU AI Act; and the conformance requirements an implementation SHALL meet to claim alignment with this specification.

1.2 Applicable AI systems

This document applies to AI systems that generate outputs consumed by humans or by downstream automated processes, in particular generative and decision-support systems deployed in high-risk contexts as defined by applicable law. It is technology-neutral as to model architecture.

1.3 Out of scope

This document does not specify model training, model architecture, or the internal parameters of any AI system; it concerns only the external verification and control of outputs. It does not constitute legal advice, and conformance to it does not by itself establish legal compliance with any regulation.

1.4 Intended users

The intended users are standards bodies and their technical committees; regulators and conformity-assessment bodies; AI providers and deployers seeking a technical basis for the obligations they face; and independent auditors.

1.5 Normative conventions

In this document SHALL indicates a requirement, SHOULD a recommendation, MAY a permission, and CAN a capability. These verbs define conformance to this document only (Clause 11); as stated in the Note on Method, they impose no legal obligation and presume no adoption by any standards body. References to clauses and

annexes are to this document unless otherwise stated.

2. Normative References

The following instruments are referred to in this document. They are real, externally maintained artifacts; HNS does not control them and claims no endorsement by them.

2.1 ISO/IEC JTC 1/SC 42 — Artificial Intelligence

Reference	Title / subject
ISO/IEC 42001:2023	AI management system (AIMS) — requirements
ISO/IEC 42005:2025	AI system impact assessment
ISO/IEC 42006	Requirements for bodies auditing and certifying AIMS
ISO/IEC 22989:2022	AI concepts and terminology
ISO/IEC 23894:2023	AI risk management
ISO/IEC 5338	AI system life cycle processes
ISO/IEC TR 5469	Functional safety and AI systems
ISO/IEC 24029-2	Robustness of neural networks — formal methods
ISO/IEC 38507:2022	Governance implications of the use of AI by organizations

2.2 European Union regulatory framework

Reference	Subject
Regulation (EU) 2024/1689	Artificial Intelligence Act (entered into force Aug 2024)
Digital Omnibus on AI (2026)	Provisional agreement of 7 May 2026 amending AI Act deadlines
AI Act Articles 9–15, 17	High-risk requirements; record-keeping; transparency; human oversight; QMS
AI Act Annex III	High-risk use-case categories
Directive (EU) 2022/2555 (NIS2)	Network and information security
CoE CETS No. 225	Council of Europe Framework Convention on AI

2.3 W3C, IEEE, NIST, ITU-T, and CEN-CENELEC

Reference	Subject
W3C PROV-O; RDF; JSON-LD; SPARQL	Provenance ontology and the linked-data stack for audit records
NIST AI RMF 1.0 (2023)	AI Risk Management Framework
NIST-AI-600-1 (2024)	Generative AI Profile
IEEE 7001; IEEE P2863	Transparency; organizational governance of AI

Reference	Subject
IEC 61508; ISO/IEC 27001	Functional safety; information security management
ISO/IEC/IEEE 29119	Software testing (conformance test basis)
CEN-CENELEC JTC 21 (prEN 18228, 18284, 18283)	Draft harmonised standards for the EU AI Act
ITU-T (AI for telecommunications)	External verification requirements in telecom AI

2.4 How these references are used

The referenced instruments play three distinct roles, and keeping the roles separate matters. The ISO/IEC SC 42 family and the governance standards supply the management and life-cycle context into which the HNS technical layer fits. The EU AI Act, NIS2, and the Council of Europe Convention supply the binding obligations the layer is designed to evidence. The linked-data, functional-safety, security, and testing standards supply the engineering substrate — the formats, integrity, and test basis — on which the layer is built. A reference’s role determines how this document treats it: context is aligned with, obligation is evidenced, substrate is conformed to.

2A. Legal Basis for Referenced Standards

2A.1 Legal hierarchy

The instruments above occupy different rungs of bindingness, and this specification keeps them distinct. Binding law (the AI Act; NIS2) imposes obligations directly. Harmonised standards (the CEN-CENELEC drafts), once cited in the Official Journal, confer a presumption of conformity with the law but are not themselves law. International standards (the ISO/IEC family) and framework documents (the NIST RMF) are voluntary unless incorporated by reference into law or contract. HNS, as proposed here, is below all of these: a candidate technical specification that would acquire force only through adoption by one of them.

2A.2 The conformity-of-presumption chain

The mechanism this volume targets is the European one, in which a harmonised standard cited in the Official Journal gives a manufacturer a presumption of conformity with the corresponding legal requirement. The proposal is that an HNS-based technical specification, adopted as or referenced by such a standard, would convert the AI Act’s high-risk requirements (Articles 9–15) into checkable, inference-level criteria. Whether any standard confers presumption of conformity is a decision of the European Commission, not of its drafters; this is stated to avoid overclaiming.

2A.3 HNS component legal-basis matrix

HNS component	Primary legal / standards hook	Nature of the hook
HNS-36/144/864 (Vols 2–3)	AI Act Art. 15 (accuracy, robustness)	proposed technical means
SMS-6 (Vol 4)	AI Act Art. 9 (risk); CoE CETS 225	proposed technical means
EVA (Vol 5)	AI Act Art. 12 (logging), Art. 13 (transparency)	proposed technical means
ECS (Clause 9)	AI Act Art. 14 (human oversight)	proposed technical means
Integrated stack (Clause 10)	ISO/IEC 42001 §5; ISO/IEC 38507	proposed technical means

2A.4 The limit of the chain

The presumption-of-conformity chain has a limit this specification states rather than glosses. A technical specification supports a legal requirement only once a recognized body adopts it and, in the European case, only once its reference is published in the Official Journal; until then the mapping is aspirational. Moreover, presumption of conformity is rebuttable: it shifts the burden of proof, it does not settle the question. HNS is therefore best understood as a candidate input to that chain, whose value is realized only through the standards process set out in Clause 15.

2B. JSON-LD, RDF, and the HNS Ontology

2B.1 The linked-data stack

Conformance evidence under this specification is machine-readable by construction. Records are expressed in JSON-LD, interpreted as RDF, conformant to the W3C PROV-O provenance ontology, and queryable in SPARQL. This choice means that an audit record is not a proprietary log but a graph any third party can parse, relate, and query with standard tooling — the property that makes independent verification practical.

2B.2 The HNS namespace

HNS terms are published under a stable namespace prefix (here written `hns:`) so that each coordinate, operator, and verdict is a dereferenceable term rather than an opaque string. A cell, an operator, a grounding layer, and an audit field each have a defined IRI.

2B.3 Cell definition in JSON-LD

A coordinate cell is defined as a typed entity carrying its layer, category, quadrant, and operator:

```
{
  "@context": "https://naturalstructureworks.com/hns/context",
  "@id": "hns:cell/L6-C5-Q3-02",
  "@type": "hns:Coordinate",
  "hns:layer": "L6", "hns:category": "C5",
  "hns:quadrant": "Q3", "hns:operator": "02"
}
```

2B.4 EVA audit record in JSON-LD / PROV-O

Each verification step is recorded as a PROV-O entity bound to the coordinate and the verdict (the schema is normative in Annex B):

```
{
  "@context": "https://www.w3.org/ns/prov",
  "@type": "prov:Entity",
  "eva:step": 42,
  "eva:configuration": "L6 x C5 x Q3 . 02",
  "eva:assertedOperator": "02",
  "eva:supportedOperator": "01",
  "eva:groundingScore": 0.46,
  "eva:attenuation": 0.46,
  "eva:flag": "UnsupportedCausality/confounding",
  "eva:ecsAction": "GATE_HOLD",
}
```

```
"prov:wasGeneratedBy": "eva:sidecar",  
"eva:signature": "tpm:sha256:9f2c..."  
}
```

2B.5 SPARQL query examples

Because records are RDF, conformance questions become queries.

Two examples:

```
# Query 1 - all records that failed verification  
SELECT ?step ?config ?flag WHERE {  
  ?r eva:flag ?flag ; eva:step ?step ;  
    eva:configuration ?config .  
  FILTER(?flag != "none")  
}  
  
# Query 2 - session risk profile by natural layer  
SELECT ?layer (AVG(?s) AS ?meanGrounding) WHERE {  
  ?r eva:configuration ?config ; eva:groundingScore ?s .  
  BIND(SUBSTR(?config,1,2) AS ?layer)  
} GROUP BY ?layer
```

3. Terms and Definitions

coordinate — the position of a claim in the HNS-36/144/864 system (Volumes 2–3).

grounding score (s) — the composite verdict of the Social Meta Layers, $s \in [0,1]$ (Volume 4).

attenuation — the pre-decode scaling $g(s)$ applied to a candidate output (Volume 3, 06).

EVA — External Verification Architecture: the external, tamper-evident recording layer (Volume 5).

ECS — External Control System: the external control gate and emergency override defined in Clause 9.

audit record — a JSON-LD / PROV-O entity evidencing the verification of one inference step.

conformance — satisfaction of the requirements of Clause 11 of this document.

presumption of conformity — the legal effect, under EU law, of a harmonised standard cited in the Official Journal.

4. HNS Overview (Normative Recap)

This clause recapitulates the HNS components only to the depth this specification needs. The normative source for each is the volume cited; in case of conflict, that volume and the Unified Structural Coordinate Table govern.

4.1 The Multi-Axis Verification process

Reliability is treated as the conjunction of three axes: Axis 1 (generative) proposes; Axis 2 (structural / causal) positions and audits; Axis 3 (grounded / executive) grounds. EVA records the result and ECS gates the output. The axes are approximately independent, which is why their conjunction adds assurance none provides alone.

4.2 The six Natural Layers (canonical)

ID	Name	Lineage
L1	Physical / Substrate	Physics / BFO
L2	Cellular / Informational	Computational neuroscience
L3	Cognitive / System 1	Kahneman Type 1
L4	Operational / System 2	Kahneman Type 2
L5	Micro-Social / Interaction	Habermas
L6	Macro-Social / Institutional	Institutional theory / BFO

4.3 The six Abstract Categories (canonical)

ID	Name	Lineage
C1	Continuant / Entity	Aristotle / BFO
C2	Occurrent / Process	Aristotle / BFO
C3	Relational / Boundary	Aristotle / Kant
C4	Quality / State	Kant
C5	Transformation / Operation	Pearl (do-operator)
C6	Meta-Level / Schema	Kant

NOTE *Naming reconciliation. Earlier drafts of this specification used different layer and category names (e.g. “Substrate / Process...” for the categories). The canonical names above, fixed in the Unified Structural Coordinate Table and Volume 2, are authoritative and supersede those drafts.*

5. HNS-36, HNS-144, and HNS-864 (Normative Recap)

5.1 HNS-36 and HNS-144

HNS-36 is the product of the six Natural Layers and six Abstract Categories (36 base cells); HNS-144 extends it by four Onto-Modal Quadrants (Subjective/Objective $\times$ Necessary/Contingent). The coordinate positions a claim and observes its modal stance, suppressing Layer Jump, Category Ambiguity, and Metaphor Contamination. The full specification is Volume 2; Annex A reproduces the normative 36-cell reference.

5.2 HNS-864 and the structural risk score

HNS-864 extends HNS-144 by six Inference & Control Operators (O1–O3, Pearl’s hierarchy; O4–O5, perceptual control; O6, attenuation). The analytical engine computes, for each candidate, the operator asserted against the operator the evidence supports; a mismatch drives a structural risk score, and the grounding score s drives the attenuation $O6 = g(s)$. The full specification is Volume 3.

$$output = g(s) \cdot raw_output, \quad g(0) = 0, \quad g \text{ monotone increasing}, \quad g(s) \in [0,1]$$

5.3 SMS-6 grounding

The six Social Meta Layers (Volume 4) ground the output from the somatic base to the universal scale; their conjunction is the grounding score s consumed above. SMS-5 (Governance) and SMS-6 (Universal) are the layers most directly engaged by this volume’s technical mapping and conformance criteria..

8. EVA — External Verification Architecture

8.1 Role in this specification

EVA (Volume 5) is the recording and verification layer. It satisfies three structural conditions — Verifiability, Transparency, Physical Immutability — by a sidecar that records each step in JSON-LD/PROV-O and a hardware root of trust that signs the record. In this specification EVA is the source of the audit record against which conformance is judged. EVA records and proves; it does not, by itself, stop output.

8.2 Mandatory audit-record fields

An implementation claiming conformance SHALL emit, for each verified step, at least the following fields (schema normative in Annex B): the coordinate configuration; the asserted and supported operators; the grounding score; the attenuation applied; any failure flag and sub-form; the per-layer SMS verdicts; the generating activity; and a hardware signature.

8.3 Mapping to EU AI Act Articles 12 and 13

AI Act provision	EVA realization
Art. 12 — record-keeping / logging	The JSON-LD audit record gives the required log a schema and a granularity (per inference step)
Art. 13 — transparency / information to deployers	The third-party-readable, tamper-evident record makes the system’s behaviour inspectable
Art. 15 — accuracy, robustness	Operator-mismatch and grounding scores provide an inference-level robustness signal

8.4 The boundary of EVA

EVA’s power is epistemic, and its boundary should be stated so that Clause 9 is not misread as part of it. EVA records faithfully, signs immutably, and exposes to third parties; it does not stop output, does not correct the model, and does not guarantee that a faithfully recorded verdict is correct — the baseline-correctness limit of Volume 5 carries into this specification. The control that acts on EVA’s verdict is the separate tier of Clause 9; conflating the recorder with the controller would overstate what either does.

9. ECS — External Control System

Where EVA records and proves, the External Control System acts. ECS is the control tier of the stack: an external gate on output delivery, with an emergency override, mapped to the human-oversight requirement of EU AI Act Article 14. It is introduced here, in Volume 6, as a component distinct from — and downstream of — the verification of Volume 5..

9.1 The three enforcement tiers

The series distinguishes three mechanisms, complementary not redundant. First, attenuation (Volume 3, O6): a pre-decode, graded, “soft” suppression that shapes what is generated. Second, EVA (Volume 5): faithful, immutable recording that proves what happened but does not stop it. Third, ECS (this clause): an external, “hard” gate that can hold or halt delivery and place a human in the loop. A complete enforcement story uses all three; ECS is the backstop the other two deliberately leave open.

9.2 The gate requirement

Under this specification, no AI output in a high-risk deployment SHOULD be delivered to a user without passing through the control gate, and in the strictest conformance level (Clause 11) SHALL NOT. The gate consumes the EVA verdict for the step: a record below the deployment’s grounding or risk threshold is held, routed for human review, or blocked, and the action taken (for example GATE_HOLD) is itself written to the record.

9.3 Mapping to EU AI Act Article 14

Art. 14 requirement	ECS realization (proposed)
14(1) — human oversight designed in	The control gate is part of the output pipeline by architecture, not an add-on
14(4)(a) — understand capacities and limits	The held record carries the coordinate and the reason for the hold
14(4)(d) — ability to intervene / interrupt	The emergency override halts delivery and requires authenticated reactivation
14(4)(e) — ability to stop the system	The override operates independently of the model’s software state

9.4 Emergency override

The emergency override SHALL operate independently of the AI system’s software state; SHOULD take effect within the latency budget set in the deployment specification (a target of ≤ 500 ms is recommended); SHALL generate a mandatory incident record within 60 seconds of activation; and SHALL require operator authentication before reactivation. A failure-mode analysis SHOULD demonstrate, to the functional-safety expectations of IEC 61508, that no single software failure can disable the override.

NOTE *Status and caution. ECS as specified — a mandatory gate and an independent override — is the author’s proposal, not an adopted requirement, and the latency and single-failure properties are engineering targets, not demonstrated results. An automated external stop is itself a safety-critical component (IEC 61508); “physical enforcement” should be read as a design goal subject to validation, not as a proven guarantee. Volume 5 deliberately placed ECS out of EVA’s scope; this clause supplies it as the separate control tier.*

9.5 Fail-safe design

An external control gate must decide what to do when it is itself uncertain or degraded, and the choice is a safety-design decision with no universally correct answer. A fail-closed gate blocks output when verification is unavailable, favouring safety over availability — appropriate where a wrong output is worse than none. A fail-open gate releases output when the gate is degraded, favouring availability — appropriate where withholding is itself harmful, as in some safety-of-life settings. The deployment specification SHALL state which mode applies per output class, and the chosen mode SHALL be recorded; the human-oversight provisions of Clause 9.3 exist precisely so that this decision is not left implicitly to the machine.

10. The Integrated HNS Stack

The integrated HNS stack is the meta-architecture that composes the coordinate (HNS-36/144/864), the grounding stack (SMS-6), the verification layer (EVA), and the control system (ECS) into one runtime. It engages the governance expectations of ISO/IEC 38507, ISO/IEC 42001 §5, the life-cycle processes of ISO/IEC 5338, and the human-oversight provision of CoE CETS 225.

NOTE *Naming. Earlier drafts called this meta-architecture “HumanOS.” This specification uses “the integrated HNS stack” for the same construct, to avoid introducing a top-level component not present in the Unified Structural Coordinate Table; “HumanOS” is retained only as a legacy name.*

10.1 Runtime data flow

Interface	From	To	Data
I-1	AI model	Axis 2 / Axis 3	candidate output stream
I-2	Axis 2 / Axis 3	EVA	coordinate, operators, grounding score
I-3	EVA	ECS	signed verdict for the step
I-4	ECS	user / deployer	released, held, or blocked output

The flow is a loop: the model proposes; Axes 2 and 3 position and ground; EVA records and signs; ECS gates; and the action returns to the record. Each interface is external to the model in the sense Volume 5 makes precise, and each is auditable after the fact.

11. Conformance Requirements

An implementation claiming conformance to this specification SHALL satisfy the requirements of this clause at the level it declares.

11.1 HNS-36 baseline

The implementation SHALL assign every verified output a coordinate in the canonical HNS-36 system (Annex A), and SHALL record the assignment. Use of HNS-144 and HNS-864 resolution is RECOMMENDED for high-risk deployments.

11.2 EVA logging

The implementation SHALL emit, for each verified step, an audit record conformant to the Annex B schema, in JSON-LD/PROV-O, and SHALL root the integrity of the record in a hardware signature. Records SHALL be retained and made available to accredited third parties.

11.3 ECS safety boundaries

Where the declared level requires it, the implementation SHALL route each output through the control gate and SHALL provide an emergency override meeting Clause 9.4. The boundaries of automated action (hold, review, block) SHALL be configured in the deployment specification and recorded.

11.4 Conformance levels

Level	Requirements	Indicative use
L1 — Recorded	HNS-36 baseline + EVA logging (Verifiability)	internal / development
L2 — Independent	+ Transparency + hardware-signed records	most high-risk deployments
L3 — Controlled	+ ECS gate and emergency override	safety-critical / public-sector

11.5 Conformance test suite

Conformance SHOULD be demonstrated through a documented test suite structured per ISO/IEC/IEEE 29119, exercising representative inputs across the coordinate and verifying that records, gating, and

override behave as specified. The suite and its results form part of the conformance evidence.

11.6 Conformance statement

An implementation claiming conformance SHALL publish a conformance statement recording: the declared level (L1–L3); which of the three conditions are met and by what means; the substrate of the hardware root of trust; the fail-safe mode per output class; the retention and third-party-access policy for records; and the version of this specification and of the Unified Structural Coordinate Table against which conformance is claimed. The statement is itself part of the audit evidence and SHALL be made available to accredited third parties.

12. Implementation Guidance

12.1 A staged adoption path

Adoption is expected to be incremental. A practical path: begin at Level 1 with coordinate assignment and JSON-LD logging in a sidecar; add hardware-signed records and process separation to reach Level 2; and add the control gate and override to reach Level 3 for the highest-stakes deployments. Each stage produces evidence usable for the next.

12.2 Proof-of-concept procedure

A minimal proof of concept assigns the HNS-36 coordinate to a stream of representative outputs, records the verdicts, and demonstrates that a third party can, from the records alone, identify the coordinate of any flagged output and confirm whether a subsequent intervention removed the flag. This is the smallest demonstration that exercises the verification-and-correction cycle the framework rests on.

12.3 A worked conformance walkthrough (informative)

To see the layers act together, follow one output. A high-risk decision-support system proposes the continuation “the applicant was rejected because of the neighbourhood.” Axis 2 positions the claim at ($L6 \times C5 \times Q3$) and reads it as asserting intervention (O2); the evidence supports only association (O1). Axis 3 returns a low governance and economic grounding, since a protected-proxy attribution is in play, and the grounding score falls to $s = 0.4$. The attenuation O6 damps the causal continuation. EVA writes the record — configuration, asserted O2, supported O1, $s = 0.4$, flag `UnsupportedCausality` — with a hardware signature. ECS, at Level 3, reads the sub-threshold record and issues `GATE_HOLD`, routing the decision to a human reviewer and writing the action to the record. A third party later queries the session (Clause 2B.5) and confirms both that the deviation was caught and that the held decision was reviewed. A single output exercises every requirement of Clause 11.

13. Security and Safety Considerations

13.1 Structural failure taxonomy

The framework names a small set of structural failures it is built to surface: Layer Jump (illegitimate transfer across levels), Category Ambiguity (conflated categories or normative kinds), Metaphor Contamination (a figure read as a mechanism), Scope Drift (unannounced change of scope), and Unsupported Causality (a causal claim above its supported operator). Each maps to a coordinate signature and to a record flag, which is what makes them externally auditable rather than matters of judgement.

13.2 Relationship to ISO/IEC 27001, IEC 61508, and NIS2

The verification and control layer is itself an asset to be secured. The integrity of records and the availability of the control gate fall within an information-security management system (ISO/IEC 27001); the emergency override is a functional-safety function (IEC 61508); and, for operators of essential services, both fall within the security obligations of NIS2. This specification assumes, and does not replace, those regimes.

13.3 Threat model for the verification layer

The verification layer is itself a target, and a specification that ignored this would be naive. Three threats are primary. Record forgery — an attacker rewrites or fabricates audit records — is countered by the hardware signature and process separation (Volume 5's Physical Immutability and Transparency). Gate bypass — output routed around the control gate — is countered by making the gate part of the delivery pipeline by architecture (Clause 9.2) rather than an optional filter. Baseline substitution — the reference baseline silently replaced with a permissive one — is countered by binding the conformance statement to a named, versioned baseline (Clause 11.6) a third party can check. None of these countermeasures is absolute; each raises the cost of an attack and, more importantly, makes a successful attack leave a detectable trace.

14. Ethical and Societal Considerations

Three considerations bear stating. First, externality is a safeguard against capture: a verification controlled by the verified party is no safeguard, which is why both the recorder and the reference baseline must be independent and public. Second, recording is not neutral: an inference-level record is a powerful artifact, and its retention, access, and privacy must be governed so that a safety mechanism does not become a surveillance one. Third, the control gate carries its own risk: an automated stop can fail open or fail closed, and the human-oversight provisions exist precisely so that the final judgement remains with a person. The framework is offered in the belief that provable, externally checkable safety serves the public better than unverifiable assurance — a belief, not a proof.

15. Standardization Status and Pathway

15.1 Current status

HNS is not, at the time of writing, part of any adopted standard or law. It is a published, version-controlled framework with a draft New Work Item prepared for submission to ISO/IEC JTC 1/SC 42. The mappings in this volume are proposals for how HNS would support the cited instruments if adopted, not descriptions of existing requirements.

15.2 The regulatory window

The timing is favourable to the proposal and is stated plainly, not triumphantly. The EU AI Act's high-risk obligations have been deferred — under the May 2026 Digital Omnibus agreement, Annex III obligations move to December 2027 — with the deferral linked to the availability of harmonised standards and support tools. The harmonised standards themselves have slipped from their 2025 deadline toward 2026, and the regulator has acknowledged asking for standardization in areas where no settled technical state of the art exists. That is the opening this specification addresses: a defined need for an inference-level technical layer that does not yet have an agreed form.

15.3 Pathway

The pathway is the one set out in Volume 5: publish as a citable preprint; submit a New Work Item to SC 42 through the national body; align with CEN-CENELEC for the EU AI Act; build an implementation track record through proofs of concept; and transfer governance to a neutral multi-stakeholder body so that the standard's independence outlasts its author.

15.4 Limitations

The limitations are real and are not hidden. HNS has not been reviewed by any standards body, and adoption is uncertain and not within the author's gift. The baseline-correctness problem of Volume 5 stands: the stack records and enforces faithfully, but the correctness of its verdicts depends on the correctness of the baseline, which is

itself revisable. The feasibility of per-step verification and gating within a latency budget is an engineering claim to be demonstrated. And the enforcement the framework offers is conditional on deployment: a provider who controls the whole stack can decline to adopt it, which is exactly why the standards and conformance verification route, rather than the purely technical one, is the subject of this volume..

Annex A — HNS-36 Cell Reference (Normative)

The 36 base cells are the product of the six Natural Layers (rows) and six Abstract Categories (columns). Cell identifiers are of the form L×C. The full per-cell specification is Volume 2; this annex fixes the canonical grid.

	C1	C2	C3	C4	C5	C6
L1	L1C1	L1C2	L1C3	L1C4	L1C5	L1C6
L2	L2C1	L2C2	L2C3	L2C4	L2C5	L2C6
L3	L3C1	L3C2	L3C3	L3C4	L3C5	L3C6
L4	L4C1	L4C2	L4C3	L4C4	L4C5	L4C6
L5	L5C1	L5C2	L5C3	L5C4	L5C5	L5C6
L6	L6C1	L6C2	L6C3	L6C4	L6C5	L6C6

Columns: C1 Continuant, C2 Occurrent, C3 Relational, C4 Quality, C5 Transformation, C6 Meta-Level. Rows: L1 Physical, L2 Cellular, L3 Cognitive, L4 Operational, L5 Micro-Social, L6 Macro-Social.

Annex B — EVA Audit Record Schema (Normative)

Each verified step SHALL be recorded as a JSON-LD/PROV-O entity with at least the following fields:

- `eva:configuration` — the HNS coordinate ($L \times C \times Q \cdot O$); string.
- `eva:assertedOperator` / `eva:supportedOperator` — operators 01–06; string.
- `eva:groundingScore` — composite SMS grounding $s \in [0,1]$; number.
- `eva:attenuation` — applied factor $g(s) \in [0,1]$; number.
- `eva:flag` — failure type and sub-form, or “none”; string.
- `eva:smsVerdicts` — per-layer grounding verdicts; object.
- `eva:ecsAction` — control action taken (RELEASE / GATE_HOLD / BLOCK); string.
- `prov:wasGeneratedBy` / `eva:signature` — generating activity; hardware signature; string.

Annex C — Audit Log Examples (Informative)

A failing record (causal over-claim held by the gate):

```
{ "eva:step": 42, "eva:configuration": "L6 x C5 x Q3 . 02",  
  "eva:assertedOperator": "02", "eva:supportedOperator": "01",  
  "eva:groundingScore": 0.46, "eva:attenuation": 0.46,  
  "eva:flag": "UnsupportedCausality/confounding",  
  "eva:ecsAction": "GATE_HOLD" }
```

A passing record (released):

```
{ "eva:step": 43, "eva:configuration": "L4 x C5 x Q3 . 02",  
  "eva:assertedOperator": "02", "eva:supportedOperator": "02",  
  "eva:groundingScore": 0.95, "eva:attenuation": 0.95,  
  "eva:flag": "none", "eva:ecsAction": "RELEASE" }
```


Annex D — EU AI Act Mapping (Normative)

Proposed mapping of HNS components to the high-risk requirements of the AI Act. This is a proposal for adoption, not a statement of existing conformity.

AI Act provision	HNS component	Means
Art. 9 — risk management	SMS-6; structural failure taxonomy	inference-level risk signal
Art. 10 — data governance	coordinate assignment; provenance record	structural classification
Art. 12 — record-keeping	EVA audit record	per-step JSON-LD log
Art. 13 — transparency	EVA third-party-readable record	inspectable behaviour
Art. 14 — human oversight	ECS gate and override	hold / review / stop
Art. 15 — accuracy, robustness	operator-mismatch; grounding score	inference-level metric

NOTE *Timeline. Under the May 2026 Digital Omnibus agreement (subject to formal adoption), Annex III high-risk obligations apply from 2 December 2027 and Annex I product-embedded obligations from 2 August 2028, with the high-risk timeline linked to the availability of harmonised standards.*

D.1 Annex III high-risk categories

For reference, the AI Act’s Annex III enumerates the high-risk use-case areas to which the obligations mapped above principally apply:

- biometrics, where permitted; safety components of critical infrastructure;
- education and vocational training; employment, worker management, and access to self-employment;
- access to essential private and public services and benefits, including creditworthiness and insurance;
- law enforcement; migration, asylum, and border control; administration of justice and democratic processes.

The HNS components of this annex are proposed as a common technical means of evidencing the Article 9–15 obligations across these areas, irrespective of domain.

Annex E — Runtime Architecture (Informative)

The integrated stack composes the components of Clause 10 into a single pipeline: model → Axis 2 / Axis 3 (coordinate, operators, grounding) → attenuation → EVA (record, sign) → ECS (gate, override) → release. The sidecar and the control gate run in process spaces separate from the model; the hardware root of trust signs records and anchors the override. The architecture is the engineering expression of the three enforcement tiers of Clause 9.1.

Annex F — Glossary (Informative)

MAV — Multi-Axis Verification: the conjunction of the generative, structural, and grounded axes.

HNS-36 / 144 / 864 — the nested structural coordinate (Volumes 2-3).

SMS-6 — the six Social Meta Layers; the Axis-3 grounding stack (Volume 4).

EVA — External Verification Architecture; the recording / verification layer (Volume 5).

ECS — External Control System; the control gate and emergency override (Clause 9).

integrated HNS stack — the meta-architecture composing all of the above (legacy name: HumanOS).

presumption of conformity — the EU-law effect of a harmonised standard cited in the Official Journal.

conformance level — L1 Recorded, L2 Independent, or L3 Controlled (Clause 11.4).

Annex G — Standards × HNS Component Alignment (Normative)

A representative alignment of international instruments to HNS components. The mapping states intended support, proposed for adoption; it does not assert that any body has adopted HNS. The set is illustrative of the coverage, not exhaustive.

Instrument	HNS component	Alignment (proposed)
ISO/IEC 42001:2023 §5–10	Integrated stack; EVA	AIMS controls evidenced at inference level
ISO/IEC 42005:2025	SMS-6; coordinate	impact assessment fed by structural signals
ISO/IEC 42006	conformance (Cl. 11)	third-party certification of AIMS
ISO/IEC 22989:2022	terms (Cl. 3)	shared concepts and terminology
ISO/IEC 23894:2023	SMS-6; risk taxonomy	risk management process
ISO/IEC 5338	integrated stack	AI system life-cycle processes
ISO/IEC TR 5469	ECS	functional safety of AI systems
ISO/IEC 24029-2	HNS-864; operators	robustness assessment
ISO/IEC 38507:2022	integrated stack	governance implications
ISO/IEC 27001	EVA; ECS	security of records and control assets
IEC 61508	ECS override	functional-safety integrity
ISO/IEC/IEEE 29119	conformance (Cl. 11.5)	test-suite basis
W3C PROV-O / JSON-LD / SPARQL	EVA record (Cl. 2B)	machine-readable provenance
NIST AI RMF 1.0	MAV; SMS-6	MEASURE / MANAGE functions
NIST-AI-600-1	failure taxonomy	generative-AI risk coverage
IEEE 7001	EVA	transparency of autonomous systems
EU AI Act Art. 9–15	see Annex D	high-risk requirements
EU AI Act Art. 12 / 13	EVA	logging; transparency
EU AI Act Art. 14	ECS	human oversight
CEN-CENELEC prEN 18228 / 18284	SMS-6; coordinate	draft harmonised: risk; data quality
NIS2 (EU 2022/2555)	EVA; ECS	security obligations for essential services
CoE CETS 225	integrated stack; ECS	human-oversight convention

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